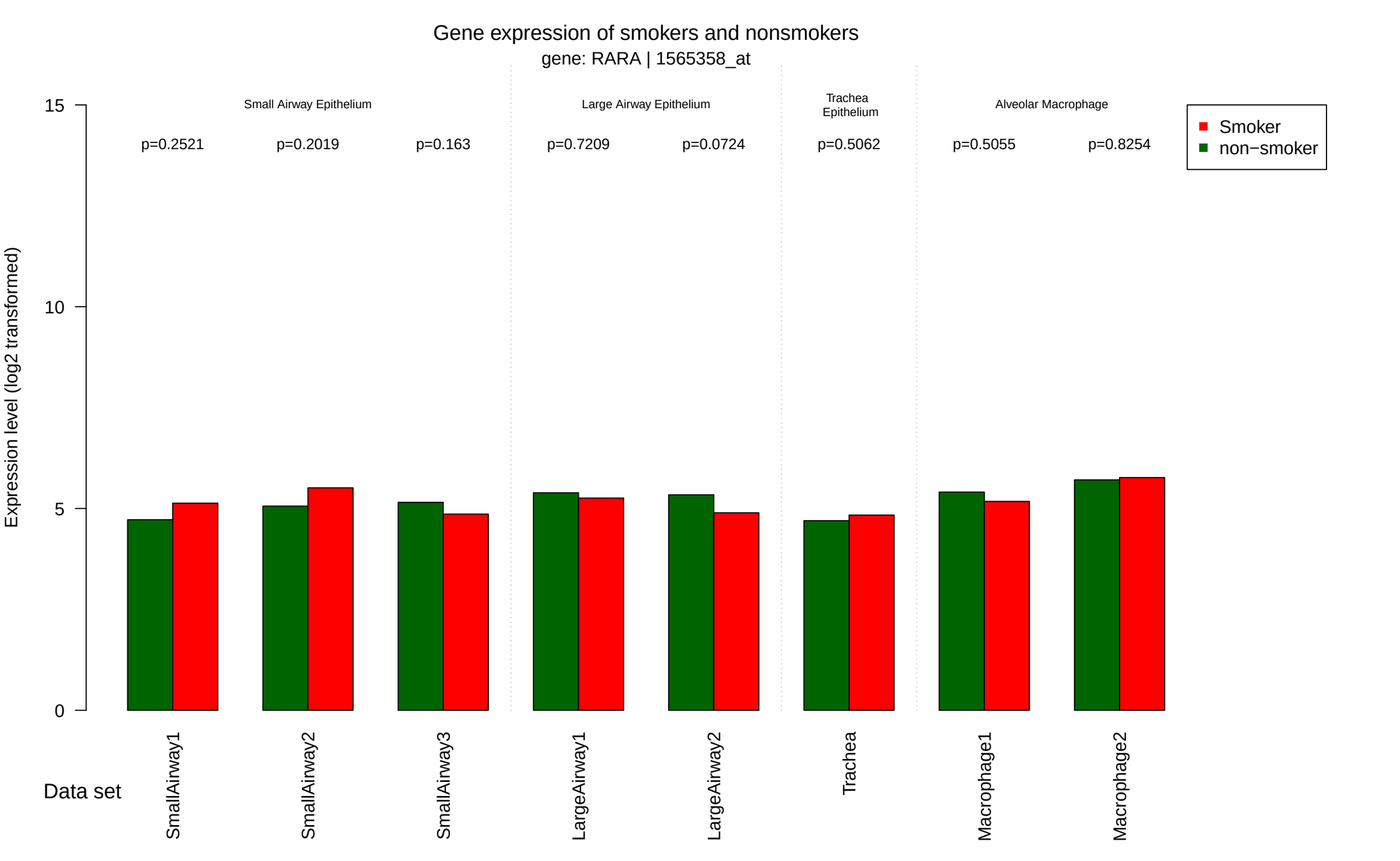
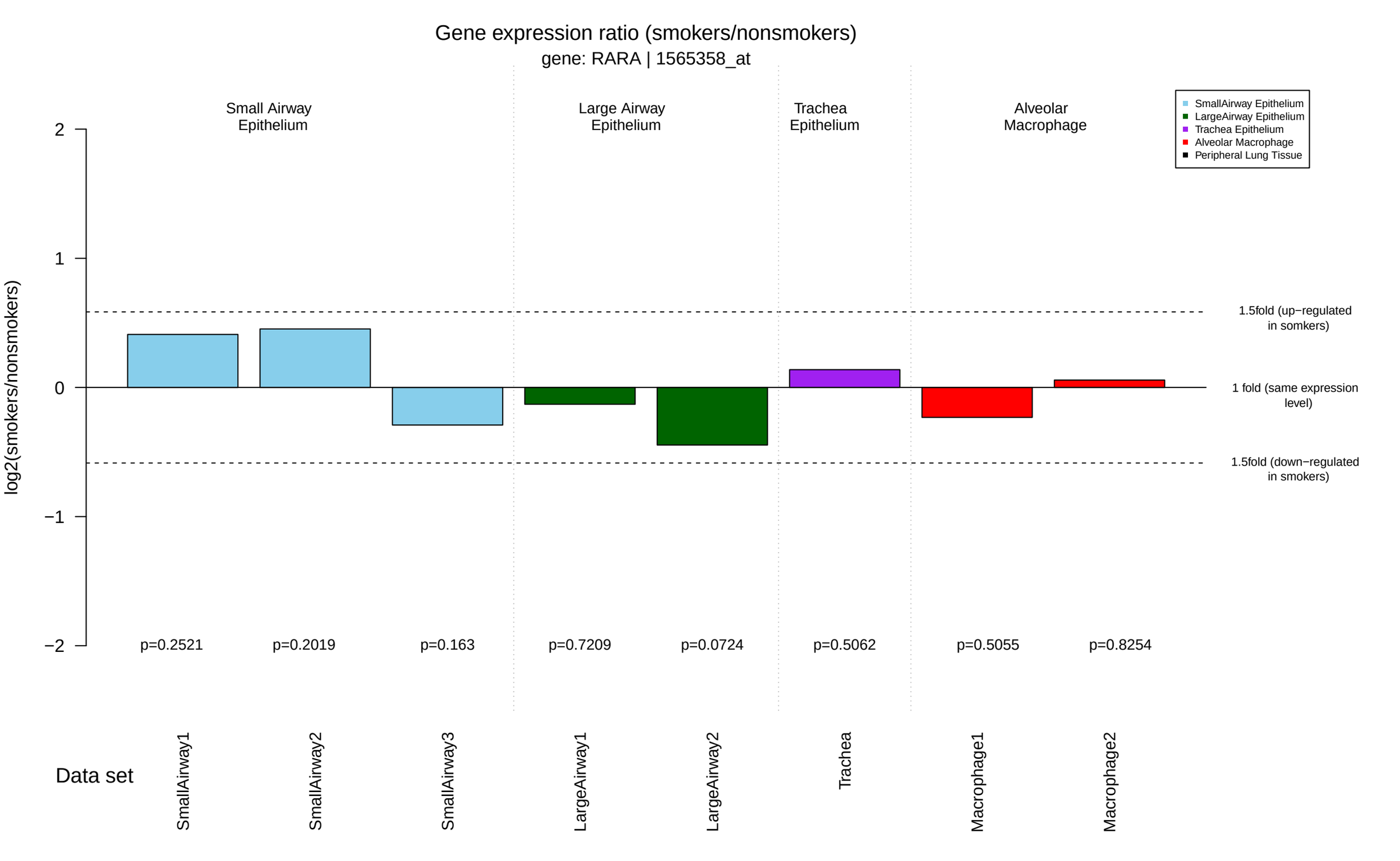
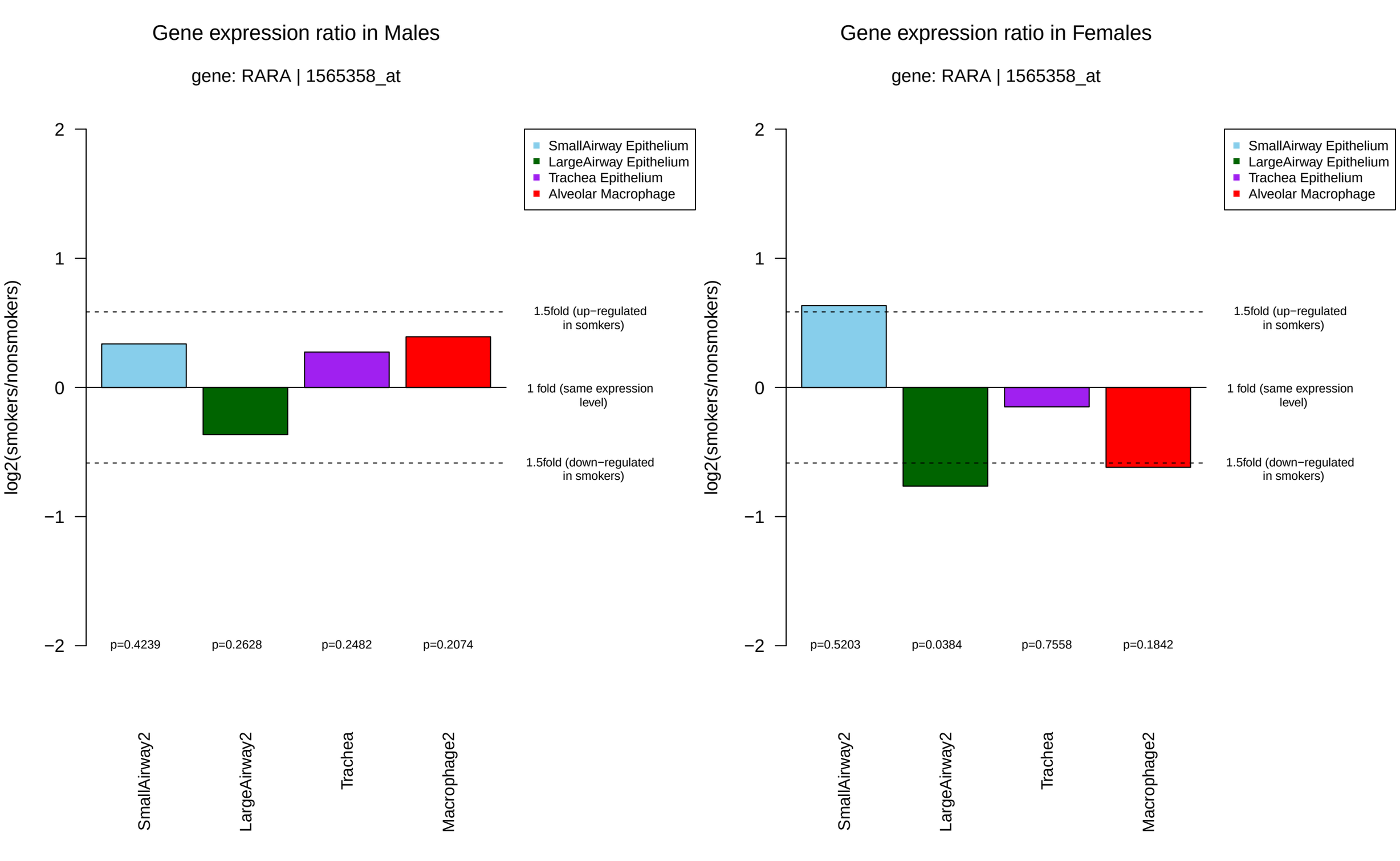
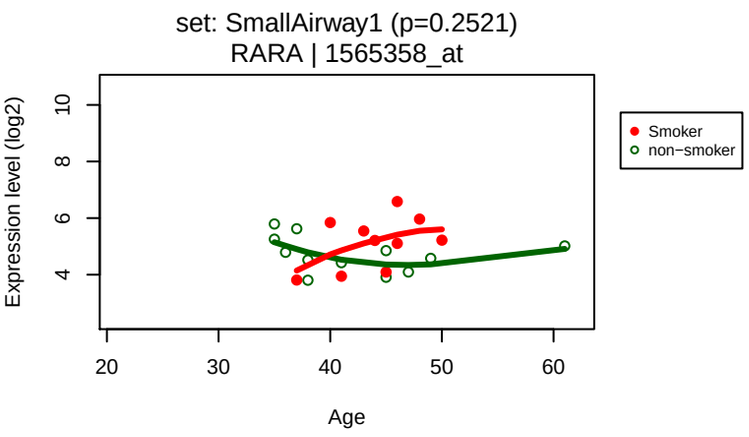
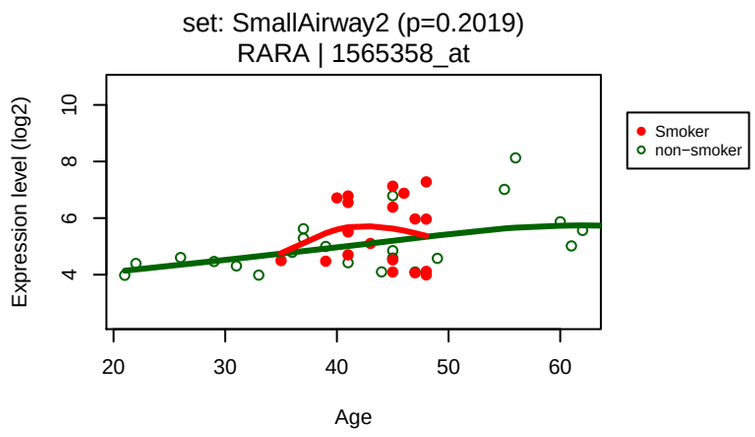
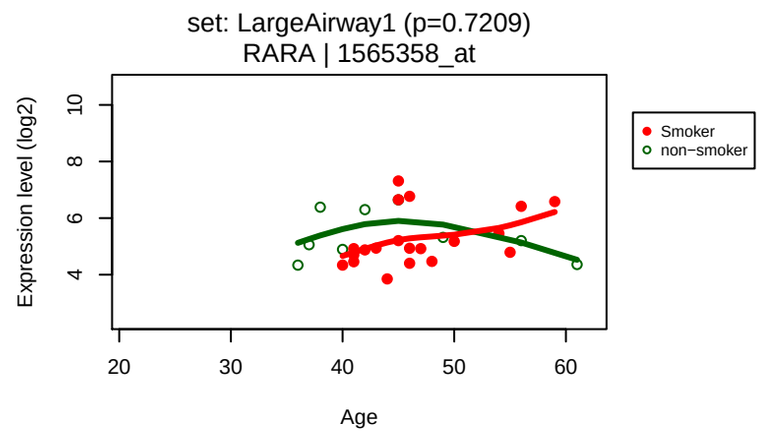

Gene expression of smokers and nonsmokers
gene: RARA | 1565358_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2521
p=0.2019
p=0.163
p=0.7209
p=0.0724
p=0.5062
p=0.5055
p=0.8254
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

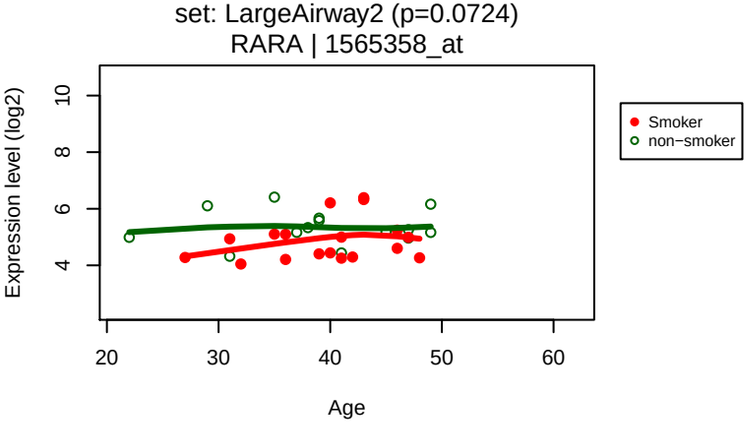

Gene expression ratio (smokers/nonsmokers)
gene: RARA | 1565358_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2521
p=0.2019
p=0.163
p=0.7209
p=0.0724
p=0.5062
p=0.5055
p=0.8254
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

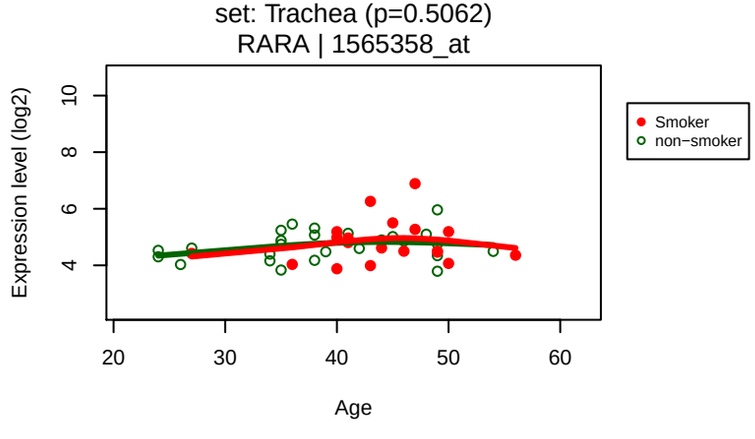

Gene expression ratio in Males
gene: RARA | 1565358_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4239
p=0.2628
p=0.2482
p=0.2074
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RARA | 1565358_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5203
p=0.0384
p=0.7558
p=0.1842
SmallAirway2
LargeAirway2
Trachea
Macrophage2

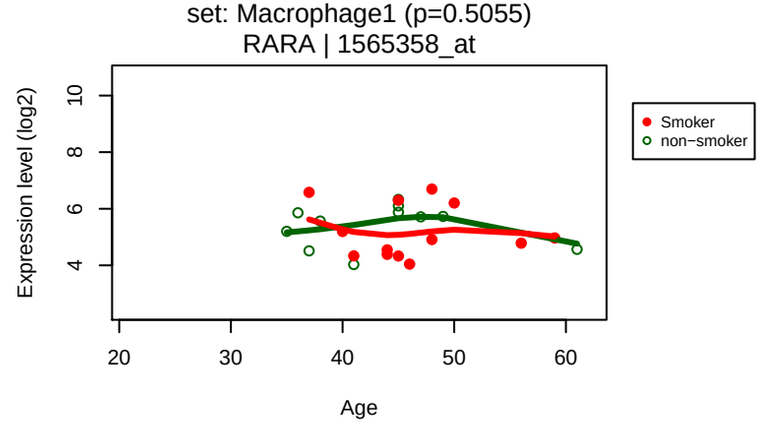

set: SmallAirway1 (p=0.2521)
RARA | 1565358_at
Expression level (log2)
Age
Smoker
non−smoker

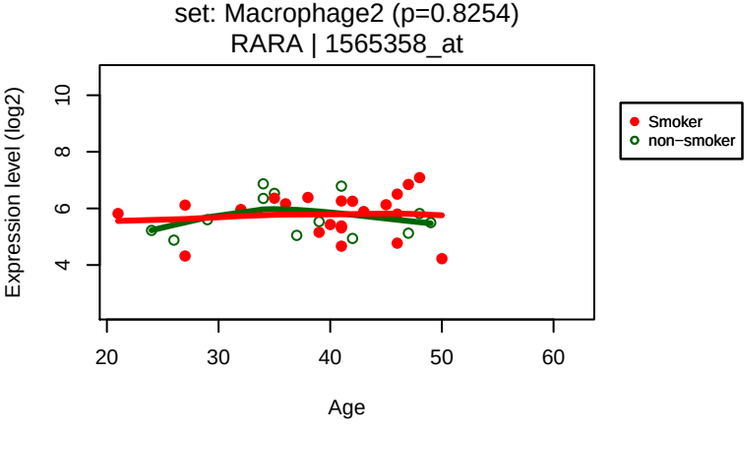

set: SmallAirway2 (p=0.2019)
RARA | 1565358_at
Expression level (log2)
Age
Smoker
non−smoker

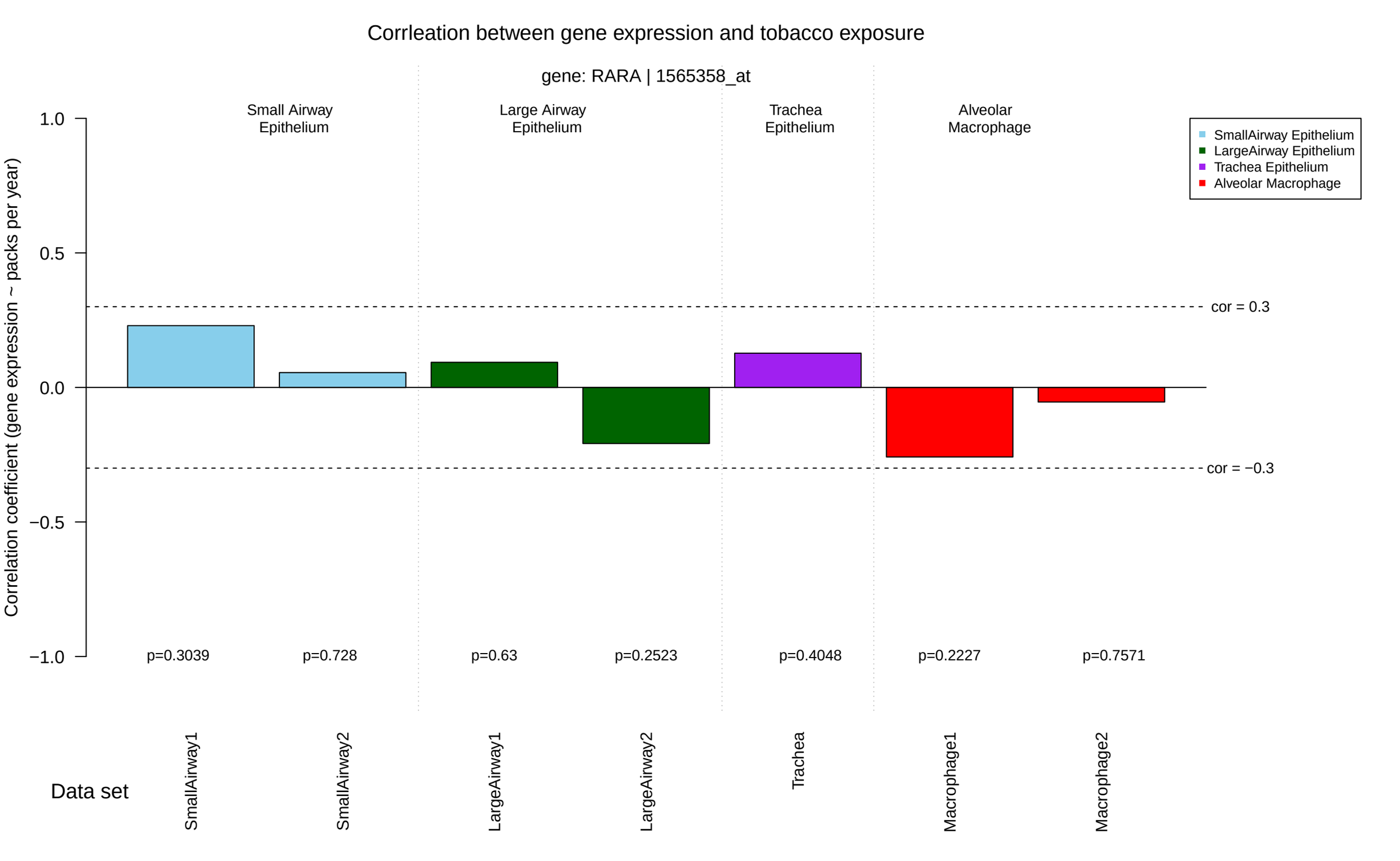

set: LargeAirway1 (p=0.7209)
RARA | 1565358_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0724)
RARA | 1565358_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5062)
RARA | 1565358_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5055)
RARA | 1565358_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8254)
RARA | 1565358_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RARA | 1565358_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3039
p=0.728
p=0.63
p=0.2523
p=0.4048
p=0.2227
p=0.7571
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2